

TOPS Technologies

Interview Questions — SE - Overview of IT Industry

Total Questions: 35

1. What is the difference between a product-based and a service-based IT company? Give an example of each.

Hint: Think about whether the company builds its own products or delivers solutions for clients.

2. Name three common career roles in the IT industry and briefly describe what each one does.
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3. How do teams use tools like GitHub, Slack, and Jira to collaborate on software projects?

Hint: Focus on what each tool helps the team achieve.

4. In an open-source GitHub project, what are some typical roles contributors might have?
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5. If you wanted to find out about the latest emerging tech jobs, how could you use an AI tool like ChatGPT to help you?
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6. What is a software application, and what are its main components?

Hint: Think about what layers or parts make up most apps.

7. Explain the three main layers in typical software architecture: UI, Business Logic, and Database.
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8. What is the difference between a desktop application, a web application, and a mobile application?
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9. Describe, step by step, what happens when you refresh your feed in a social media app from the user's device to the database and back.
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10. What is client-server communication? Give an example using a browser and a backend server.
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11. What is the Internet, and how do computers communicate over it?
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12. Briefly explain what each of these protocols is used for: HTTP, HTTPS, FTP, SMTP.
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13. What is a domain name, and how does DNS help users reach a website?

Hint: Think about how human-friendly names are linked to IP addresses.

14. What command can you use to find the IP address of a website like google.com, and what does it show you?

15. Explain the basic structure of the DNS hierarchy, including root, TLD, domain, and subdomain.

16. What is an IP address, and how is IPv4 different from IPv6?

Hint: Consider the format and length of each type.

17. When you type a URL in your browser and hit Enter, what is the typical HTTP request and response lifecycle? Mention common status codes like 200, 404, and 500.

18. How can browser DevTools help you see what happens during a network request?

19. List and briefly describe the main phases of the Software Development Lifecycle (SDLC).

20. What is the difference between the Agile and Waterfall models of software development?

Hint: Think about how work is structured and delivered in each.

21. In Scrum, what are the roles of Product Owner, Scrum Master, and Development Team?

22. Give an example of how the SDLC phases would apply to rolling out a new feature in a real-world app.

23. What is a Jira board, and how does it help teams manage their work?

Hint: Focus on how tasks are tracked visually.

24. Why is version control important in software development projects?

25. Explain these Git concepts: Repository, Commit, Branch, Merge, Push, and Pull.

26. What is the difference between forking a repository and cloning it on GitHub?

27. How does a Pull Request work on GitHub, and why is it important for collaboration?

28. Describe the steps to initialize a new Git repository, add files, make a commit, and push it to GitHub.

29. If you see a Git error message while pushing your code, how could you use an AI tool like GitHub Copilot Chat to help resolve it? What should you check before applying the suggested fix?

30. Imagine you are working with a team on a project using GitHub and Jira. How would you use these tools together to track progress and manage code changes?
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31. How does understanding the client-server communication model help you when troubleshooting issues in a web application?
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32. Suppose you are assigned as a tester in a Scrum team. Which SDLC phases will you be most involved in, and what will your main responsibilities be?
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33. If a website is not loading and you suspect a DNS issue, what steps would you take to diagnose and resolve the problem?
- Hint:** Consider checking the domain's resolution and the commands you might use.
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34. You are asked to explain to a non-technical client why their new feature request needs to go through all SDLC phases. How would you make this clear to them?
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35. What are the benefits and possible risks of relying on AI tools like ChatGPT or Copilot for technical research or troubleshooting in your IT work?

Hint: Think about accuracy, speed, and the importance of verifying results.